UNBOX STUDIOS

MODULAR WORKSTATION · v1.0.0 NATIVE

OPERATION MANUAL & SYSTEM ARCHITECTURE

Complete Reference Manual for Musicians, Producers, Sound Designers & Audio Engineers.

ZERO-ELECTRON STANDALONE | HARDWARE GPU & AUDIO PROBE | 3-COLUMN MODULAR RACK | 3rd WAVE 24-VOICE SYNTH | STUDIO CONSOLE MIXER | CV PHYSICS ROUTING

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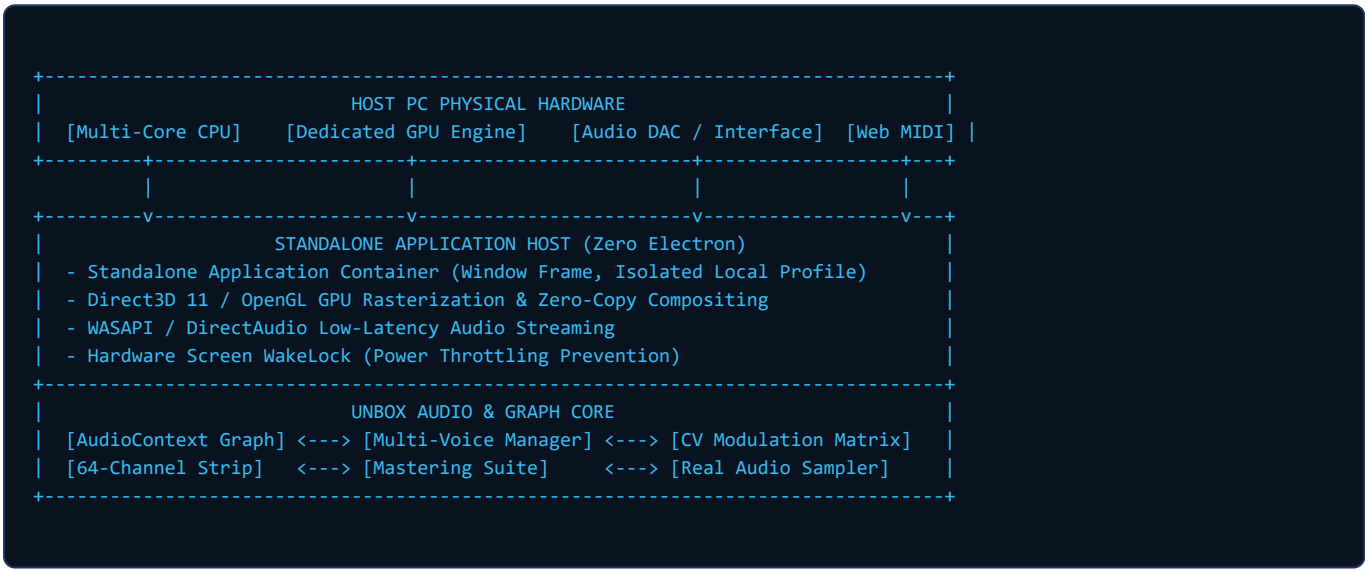
> **About This Manual**
>
> This manual provides an exhaustive operational reference for Unbox Studios. Every device parameter, CV modulation routing, audio signal path, keyboard shortcut, and hardware interaction is documented with exact values and behavior.

1. System Architecture & Standalone Runtime

Unbox Studios is a full-featured modular Digital Audio Workstation (DAW) engineered from the ground up for high fidelity, instant tactile feedback, and modular routing freedom. Combining the sonic depth of vintage rack-mounted synthesizers, outboard analog hardware emulations, and modern multi-track arrangement workflows, the application delivers studio-grade sound design capabilities directly on consumer PC hardware.

1.1 Standalone Desktop Architecture (Zero-Electron Engine)

Unbox Studios operates as a native standalone desktop application that completely bypasses the resource bloat and memory footprint of Electron. The standalone engine couples an ultra-low latency Node.js backend with your PC's native high-performance Chromium hardware renderer.



1.2 Launch Methods

- **Double-Click Launcher:** Execute `Start-Unbox-Standalone.bat` in the root directory. This launches the application with direct GPU acceleration flags and isolated user data storage in `%LOCALAPPDATA%\UnboxStudiosStandalone`.
- **Command Line:** Run `npm run standalone` from PowerShell or Command Prompt.
- **Development Mode:** Run `npm run dev` for live hot-reloading development with full Vite middleware.

> **Hardware Boost & Power Policy**
>
> When running in standalone mode, the application automatically requests an OS WakeLock to prevent Windows from clocking down CPU cores, sleeping background audio threads, or dimming displays during critical recording and mixing sessions.

2. Workspace Navigation: The Three Core Views

Unbox Studios organizes music production into three dedicated full-screen workspaces designed for instant switching with zero latency. You can toggle between views at any time using the top navigation bar or the industry-standard function keys.

Key	Workspace View	Primary Operational Purpose
`F7`	**Workstation & Sequencer**	Sound auditioning, multi-track Part selection, per-note audio clip editing, sample slicing, and arrangement timeline.
`F6`	**3-Column Modular Rack**	Full modular synth view with physical faceplates, vacuum tubes, interactive knobs, patch cables, and rear panel CV routing.
`F5`	**Mixer & Clip Editor**	SSL-style console channel strips, EQ frequency curves, dynamics, aux send routing, master meters, and clip launcher.
`Tab`	**Flip Rack (Front / Rear)**	Instantly flips the 3-column modular rack around to expose audio jacks, CV ins/outs, and physical patch cables.

2.1 Top Cabinet Title Bar

The top cabinet title bar persists across all views and provides:

- **Brand Menu:** Access project operations (New, Open, Save, Save As, Export WAV, Project Settings).
- **Workspace Selector Tabs:** One-click switching between Workstation, Rack, and Mixer.
- **Hardware Telemetry HUD:** Real-time display of active GPU renderer, CPU cores, audio interface sample rate, buffer latency, DSP load, and display FPS.
- **Global Tool Access:** Real Audio Stems browser (`Alt+S`), AI Co-Pilot (`Alt+C`), Command Search (`Ctrl+K`), and Keyboard Shortcuts (`?`).

3. The Integrated Workstation & LCD Editor

The **Workstation** view (`F7`) combines a classic Roland/Yamaha multi-timbral workstation layout with modern digital sample manipulation.

3.1 The Multi-Timbral Part List

Every instrument loaded into the project automatically occupies a numbered Part slot in the left-hand Part List:

- **MUTE Button:** Silences the part instantly without interrupting sequencer sync.
- **LINK Button:** Links secondary parts to the primary selected part so playing notes on your MIDI keyboard or clicking keys triggers lush layered sounds across multiple synthesizers simultaneously.
- **PATCH Display:** Displays the active device name (e.g., `3rd Wave Wavetable Synth`, `Big Rusty Drums`). Double-clicking opens the Patch Browser to swap instruments while preserving track patterns.
- **LEVEL Meter:** Tactile horizontal slider controlling channel gain with a real-time stereo VU meter.

3.2 The Hardware LCD Editor & Audio Clip Chopper

The central LCD screen provides deep editing of the selected instrument. It features two primary operating modes: **CLIP Mode** and **ADSR Mode**.

Automatic Live Note Tracking

When you strike any note on your MIDI keyboard or piano roll (for example note 36 / Kick on *Big Rusty Drums*), the LCD editor instantly updates its waveform display to show the exact audio sample assigned to that note. You can immediately slice, trim, and adjust its parameters without hunting through file menus.

Control	Range	Operational Function
`AUDITION`	Trigger	Plays back the currently selected note sample with current offset and loop settings.
`OFFSET`	0.00s – 10.00s	Trims the sample start point to skip silence or jump straight into the attack transient.
`DUR`	0.1b – 16.0b	Sets the maximum playback duration in musical beats or seconds.
`PITCH`	-24 to +24 st	Pitches the individual note up or down in semitones with high-quality resampling.
`SPLIT / CHOP`	16th, 8th, 1/4, 1 Bar	Automatically chops the sample across musical grid intervals for stutter and glitch beats.
`REV`	Toggle	Reverses the note audio playback buffer instantly.
`LOOP`	Toggle	Loops the audio sample continuously while the key is held down.
`F-IN / F-OUT`	0.00s – 2.00s	Applies smooth linear fade-in and exponential fade-out ramps to prevent clicks.

4. The 3-Column Modular Rack System & Patch Physics

Pressing `F6` reveals the **3-Column Modular Rack**, a full emulation of Reason-style hardware chassis with photorealistic front panels, vacuum tube thermionic glows, and interactive knobs.

4.1 Front Panel Operation

Each unit occupies standard rack height units (1U = 44px, 2U = 88px, 3U = 132px, 4U = 176px). Controls include rotary knobs, vertical faders, toggle switches, LCD displays, and visualizers. Double-clicking any knob resets it to its factory default value.

4.2 Rear Panel & Physical Patch Cables

Pressing `Tab` flips the entire rack around to the rear panel. All audio connections, sidechain inputs, gate triggers, and CV modulation lines are exposed as physical 1/4-inch jacks.

> **Dynamic Cable Physics Engine**
>
> Patch cables behave according to real-world physics formulas. As devices are dragged or resized, cables simulate gravity, tension, droop curves, and sway with full momentum. Cables are automatically color-coded (Red, Yellow, Green, Cyan, Purple) for effortless signal tracing.

Signal Type	Jack Color	Standard Voltage / Value Range	Behavior
Audio Signals	Silver Chrome Jacks	-1.0 to +1.0 (Full Scale Float)	Carries stereo or mono audio signals directly between synth outputs, effect processors, and mixer channel inputs.
CV Pitch	Yellow / Brass Jacks	1 Volt per Octave (0.0 to 1.0)	Drives oscillator pitch tracking from sequencers, arpeggiators, and matrix units.
CV Gate	Yellow / Brass Jacks	Binary Pulse (0.0 or 1.0)	Triggers note-on envelopes, pad strikes, and step sequencer triggers.
CV Modulation	Yellow / Brass Jacks	Bipolar (-1.0 to +1.0)	Continuous modulation from LFOs and envelopes to filter cutoffs, wavetable surfers, and VCA levels.

5. 3rd Wave Advanced 24-Voice Wavetable Synthesizer

The **3rd Wave** is Unbox Studios' flagship wavetable instrument, recreating the historic sound of the vintage PPG Wave 2.2/2.3 combined with modern 24-voice polyphony, analog circuit drift, dual series filters, and the iconic master **Wave Surfer**.

`NATIVE SYNTH` `4U RACK UNIT`

5.1 Oscillator Architecture (3 Oscillators Per Voice)

Each of the 24 voices contains three independent wavetable oscillators capable of running legacy PPG, modern high-resolution, analog-modeled, or sampled transient waves.

Bank	Identifier	Wave Range	Harmonic Character
PPG Legacy	`P00 – P35`	64 waveforms per table	Vintage 8-bit stepped quantization, metallic bell sweeps, robotic vocal formants, Upper Wavetable P30, and Prophet VS digital spectra (P32–P35).
Modern User	`U00 – U55`	64 waveforms per table	Anti-aliased, 96 kHz high-resolution tables with lush morphing, glassy harmonics, and cinematic textures.
Analog Core	`A00 – A06`	Static / Variable PWM	A00 Saw, A01 Square with variable PWM, A02 Triangle, A03 Pure Sine, A04 7-voice Supersaw stack, A05 White Noise, A06 Pink Noise.
Sample Slots	`S00 – S08`	Single-cycle & Transients	Acoustic kalimba, hammer strikes, gamelan transients, and choral breath.

5.2 The Master Wave Surfer & Wave Flow

The large blue **WAVE SURFER** knob sweeps the wavetable positions of all three oscillators simultaneously across a range of **-63 to +63**. The **WAVE FLOW** knob adjusts wavetable smoothing: values below 50% replicate authentic vintage 8-bit stepped transitions, while values above 50% smoothly interpolate harmonic coefficients.

5.3 6-Stage Wave Envelope

Each voice includes a dedicated 6-stage trajectory envelope with independent Time ($T_1 dots T_6$) and Position ($P_1 dots P_6$) controls. Setting **Loop Start** and **Loop End** stages creates continuous cyclical wavetable sweeping while notes are sustained.

5.4 Dual Filters in Series

- **Filter 1 (12dB State-Variable Filter - SVF):** Zero-delay digital SVF smoothly morphing between Low-Pass, Notch, High-Pass, and dedicated Band-Pass modes with Cutoff, Resonance, and Envelope amount.
- **Filter 2 (24dB 4-Pole Low-Pass Filter):** Warm resonant lowpass ladder filter equipped with a non-linear `Drive` saturation waveshaper and `Bass Comp` (Resonance Bass Compensation) to preserve thunderous low end even at self-oscillation resonance settings.

5.5 Performance, Voice Spread & Dual FX

- **Voice Pan Spread:** Distributes voices across a wide binaural stereo image (0% mono center up to 100% wide alternating spread).
- **Circuit Drift:** Introduces subtle analog-modeled pitch detuning (±2.5 cents) and cutoff deviations per voice.
- **FX 1 Processor:** Lush Multi-Tap Chorus, BBD Analog Delay, Stereo Ping-Pong Delay, Echoplex Tape Echo, Flanger, and 6-stage Phaser.
- **FX 2 Processor:** Spacious Hall Reverb, Warm Room Reverb, Superplate Reverb, and Analog Transistor Distortion.

6. Native Synthesizer Suite

In addition to the 3rd Wave, Unbox Studios includes a versatile suite of native synthesizers covering subtractive, FM, wavetable, and hybrid analog modeling.

6.1 Thunderhead (Thor) Polysonic Synth 3U

A triple-oscillator synthesizer pairing multi-oscillator detuned saws with a warm overdrive saturation stage, chorus, and dual filters. Ideal for soaring synthwave leads, rich pad chords, and energetic plucks.

6.2 Twin-Osc (Subtractor) Analog Synth 2U

The workhorse subtractive synth inspired by classic dual-oscillator analog polyphonic instruments. Features variable octave scaling, hard sync, noise generator, 24dB resonant lowpass filter, dedicated filter ADSR, and amplifier ADSR.

6.3 Aurora (Europa) Wavetable Synth 3U

A modern shapeshifting spectral wavetable synth featuring real-time wavetable morphing, harmonic multipliers, sub-oscillator reinforcement, formant peaking filter, and vacuum tube saturation.

6.4 Monobass (Monotone) Bass Synth 1U

A dedicated 1U bass powerhouse utilizing a classic 24dB transistor ladder filter with variable drive overdrive, dedicated sub-bass reinforcement, and glide portamento. Excels at 808 sub-basses, acid squelches, and driving synth basslines.

6.5 Tri-Osc Detuned Power Synth 2U

Three stacked oscillators with micro-detuning, designed for massive supersaw trance leads, stadium chord stabs, and heavy unison textures.

6.6 Fusion (Sytrus) FM / Additive Synth 2U

Multi-operator frequency modulation synthesizer capable of pure DX7 bell chimes, crystalline electric pianos, metallic percussive hits, and complex harmonic FM sweeps.

7. Traditional Indian Classical & Folk Suite

Unbox Studios features an acoustic modeling engine dedicated to regional and classical instruments with authentic microtonal pitch bending and transient response.

7.1 Bansuri Bamboo Flute 2U

Models the acoustic breath pressure, embouchure, air turbulence, and authentic *meend* (fluid microtonal sliding between notes) characteristic of North Indian classical flute music.

7.2 Classical Harmonium 2U

Recreates the multi-reed brass voicing, hand-pumped air reservoir pressure, bellows expansion, and continuous acoustic drone coupling of devotional harmoniums.

7.3 Tabla (Dayan & Bayan) 2U

Synthesizes the physical skin tension, center syahi paste loading, and resonant acoustic cavity of the wooden high-pitched Dayan (right hand) and copper/clay low-pitched Bayan (left hand). Responds dynamically to Ga, Na, Tin, Ge, and Dha strikes.

7.4 Dhol Folk Drum & Tumbi Resonator 1U

High-energy regional folk engines: the dual-headed Punjabi Dhol drum provides massive transient punch and bass slap, while the single-string Tumbi delivers high-resonance plucks with fast decay.

8. Drums & Creative Samplers

8.1 Beatbox 8 (Redrum) Drum Computer 3U

An 8-channel programmable drum computer. Features independent pitch, decay, tone, level, and pan per voice, complete with an integrated 16-step sequencer, shuffle swing timing, and accent control.

8.2 Copycat (Mimic) Creative Sampler 3U

An 8-pad granular sampler and audio slicer. Allows dropping any audio file to automatically map slices across MIDI keys, with reverse playback, start/end trimming, pitch shifting, and sample rate reduction.

8.3 Big Rusty Drums & Recorded Instrument Packs

High-resolution real acoustic sample packs recorded with close and overhead microphones, 5 velocity layers, round-robin variations, and automatic hi-hat choking (open hat cut off by closed pedal strike).

9. Studio Outboard Effects, Dynamics & Hardware Emulations

Unbox Studios integrates an arsenal of studio-grade effects, from dirty distortion units to pristine algorithmic reverbs and analog mastering processors.

9.1 Howler (Scream 4) Sound Destruction Unit 2U

Analog overdrive and saturation unit offering 10 distinct distortion algorithms (Overdrive, Distortion, Fuzz, Tube, Tape, Screamer, Warp), alongside variable Warmth even-harmonic saturation and a dual Body resonator filter.

9.2 Cathedral (RV7000) Flagship Stereo Reverb 2U

High-density algorithmic and convolution reverb with early reflection modeling, high-frequency damping, pre-delay (0–200ms), and decay times up to 12 seconds.

9.3 Repeater (Echo) Stereo Delay 2U

Stereo tape and digital delay with millisecond or tempo-synced subdivisions (1/16, 1/8, dotted, triplets), ping-pong stereo bouncing, analog tape wow & flutter, and high/low cut filters in the feedback path.

9.4 Outboard Hardware Suite

Authentic hardware faceplates and circuit modeling for industry standards:

- **SSL G-Master Bus Compressor:** Legendary VCA glue compressor with Auto-release and High-Pass sidechain filter.
- **UA 1176 Peak Limiter:** Ultra-fast FET peak limiter with all-buttons-in ("British mode") aggressive compression.
- **Teletronix LA-2A:** Optical leveling amplifier with smooth frequency-dependent tube attack and two-stage release.
- **Pultec EQP-1A:** Classic passive tube equalizer capable of simultaneous low-frequency boost and cut ("Pultec low-end trick").

10. Modulation, CV Utilities & The AI Combinator

10.1 Matrix Pattern / CV Sequencer 2U

A 32-step analog-style pattern sequencer that outputs Note CV (pitch), Gate CV (timing), and a continuous Curve CV lane for modulating filter cutoffs, panning, and effect depths in sync with tempo.

10.2 Pulsar Dual LFO & Modulation Shaper 1U

Dual low-frequency oscillators with Sine, Triangle, Saw, Square, and Random S&H waveforms, capable of tempo sync or free-running rates from 0.05 Hz to 50 Hz.

10.3 The AI Combinator & Patch Plan Engine

The Combinator wraps multiple instruments, effect inserts, and modulation units into a unified master frame with 4 assignable Macro Knobs. The integrated AI and local planning engine analyzes text prompts (e.g., "*80s synthwave lead with tape echo and sidechained reverb*") and automatically constructs, patches, and voices the entire rack.

11. Studio Console Mixer & Mastering Chain

Pressing `F5` opens the **Studio Console Mixer**, modeling an analog large-format mixing console.

11.1 Channel Strip Signal Path

- **Input Section:** Gain trim (±18 dB), Phase Invert, and steep 18dB/oct Low-Cut filter (20–400 Hz).
- **Dynamics Section:** Soft-knee compressor with threshold, ratio, attack, release, and make-up gain.
- **4-Band Parametric EQ:** Low Shelf, Low-Mid Bell, High-Mid Bell, and High Shelf with interactive SVG response curves.
- **Auxiliary Sends:** 8 independent Aux Send knobs with Pre/Post fader switching feeding Reverb and Delay return buses.
- **Fader & Pan:** High-resolution 100mm faders with peak-hold LED metering and stereo panning.

11.2 Master Section & Telemetry HUD

The master console monitors the complete acoustic output with real-time RMS, True Peak dBSPL, Phase Correlation meter (-1 to +1), and stereo Lissajous Vectorscope.

12. Arrangement Timeline, Automation & Export

12.1 Song Blocks & Timeline Editing

The arrangement timeline allows building complete compositions using reusable **Song Blocks** (Intro, Verse, Chorus, Bridge, Outro) or freeform pattern clips. Tool modes include Pointer (`Q`), Pencil (`W`), Eraser (`E`), Razor/Split (`R`), and Mute (`T`).

12.2 Multi-Lane Automation

Every knob and slider in the rack can be automated. Enabling automation recording captures manual knob twists in real time, drawing smooth Bezier curves directly beneath the sequencer track.

12.3 High-Resolution Uncompressed Stem Exporter

Export your final production in **24-bit / 32-bit Float WAV, AIFF, or FLAC**. The export engine runs on a high-speed non-realtime `OfflineAudioContext`, rendering master mixes or isolated multi-track stems with sample-accurate precision in seconds.

13. Hardware Integration & Performance Tuning

Unbox Studios directly interfaces with your computer's hardware for maximum processing power and low latency.

13.1 Hardware Probing & Multithreading

- **CPU Multi-Core Distribution:** Automatically detects logical CPU cores (`navigator.hardwareConcurrency`) and offloads audio scheduling to dedicated Web Worker timers unaffected by UI rendering.
- **Direct GPU Acceleration:** Utilizes Direct3D / WebGL 2.0 zero-copy hardware rasterization for high-refresh visualizers (60–144 Hz).
- **Audio Buffer Latency:** Probes the host audio driver (ASIO / WASAPI) and optimizes buffer sizes down to 128 samples (2.7ms latency).

13.2 Web MIDI Controller Setup

Plug in any USB or Bluetooth MIDI keyboard (Arturia, Novation, Akai, Native Instruments). Unbox Studios automatically detects the device, maps pitch bend and modulation wheels, and routes incoming note triggers to the active track with zero configuration.

14. Complete Keyboard Shortcuts Reference

Master these industry-standard keybindings to accelerate your production workflow.

Keybinding	Action / Operation	Scope
`Space`	Start / Stop Sequencer Transport Playback	Global
`F7`	Switch to Workstation & Sequencer View	Global
`F6`	Switch to 3-Column Modular Rack View	Global
`F5`	Switch to Studio Console Mixer View	Global
`Tab`	Flip Modular Rack (Toggle Front / Rear Panels)	Rack View
`Ctrl + Z` / `Cmd + Z`	Undo Last Operation	Global
`Ctrl + Y` / `Cmd + Shift + Z`	Redo Operation	Global
`Ctrl + S`	Quick Save Project to Local Storage	Global
`Ctrl + K`	Open Global Command Search Palette	Global
`Alt + S`	Open Real Audio Stems Browser Modal	Global
`Alt + C`	Toggle AI Co-Pilot & Combinator Drawer	Global
`Alt + M`	Open Full CV Modulation Matrix	Global
`F8`	Toggle MIDI Tools & Transpose Window	Global
`Q`	Select Pointer Tool	Sequencer
`W`	Select Pencil / Note Draw Tool	Sequencer
`E`	Select Eraser Tool	Sequencer
`R`	Select Razor / Split Clip Tool	Sequencer
`T`	Select Mute Tool	Sequencer
`Delete` / `Backspace`	Delete Selected Note, Clip or Device	Global
`Alt + Drag`	Duplicate Note or Audio Clip	Sequencer
`Double Click Knob`	Reset Parameter to Default Value	Rack Faceplates
`Shift + Drag Knob`	Fine-Resolution Parameter Adjustment	Rack Faceplates

UNBOX STUDIOS · END OF OFFICIAL MANUAL

Designed for high-performance audio synthesis, sample sequencing, and modular sound design.